VCS Task

Task Description:

Establish a new directory, populate it with script files, initiate an empty repository on GitHub, convert the local directory into a Git repository, and link it to GitHub for pushing the code into the repository. Perform merge, rebase, stash commands in following GitHub repo.

Solution:

1. Establishing a new Directory:

Creating new directory named Task_Day_09 as this is my 9th task:

```
GHOST@GHOST: ~/Projects/I
GHOST@GHOST:~/Projects/DevOps_Task$ ls
README.md  Task_Day_01  Task_Day_02  Task_Day_03  Task_Day_04  Task_Day_05  Task_Day_06  Task_Day_07  Task_Day_08
GHOST@GHOST:~/Projects/DevOps_Task$ mkdir Task_Day_09
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ ls
README.md  Task_Day_01  Task_Day_02  Task_Day_03  Task_Day_04  Task_Day_05  Task_Day_06  Task_Day_07  Task_Day_08  Task_Day_09
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$
```

2. Populating it with script files:

Adding some script files into the created directory:

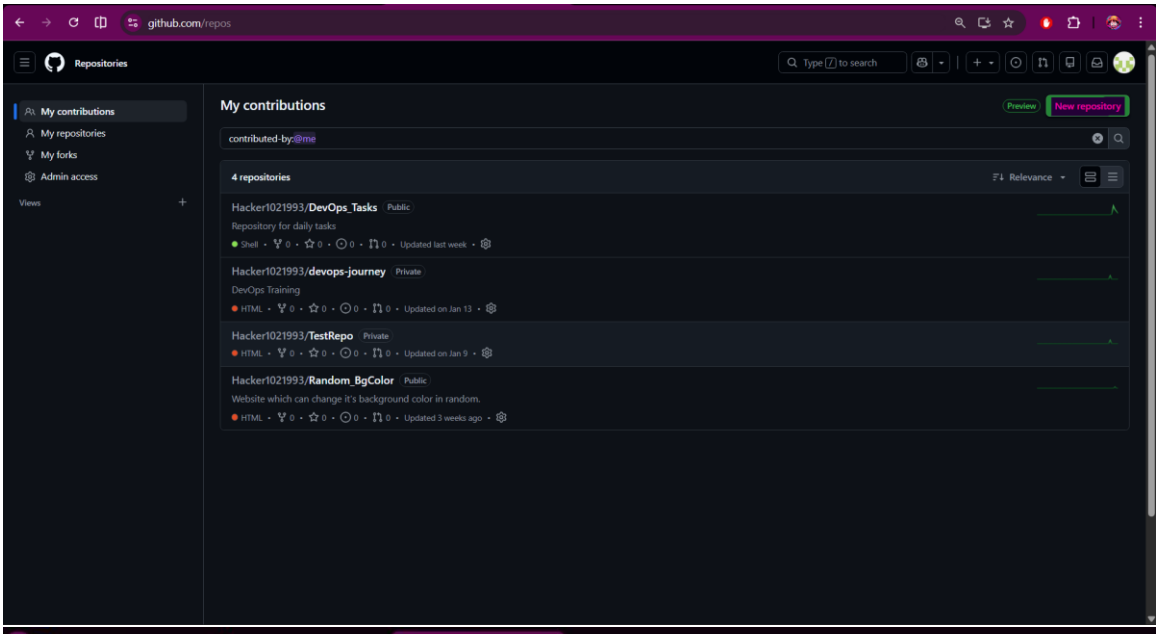
```
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ ls
README.md  Task_Day_01  Task_Day_02  Task_Day_03  Task_Day_04  Task_Day_05  Task_Day_06  Task_Day_07  Task_Day_08  Task_Day_09
GHOST@GHOST:~/Projects/DevOps_Task$ cd Task_Day_09
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ ls
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ vi calculator.sh
```

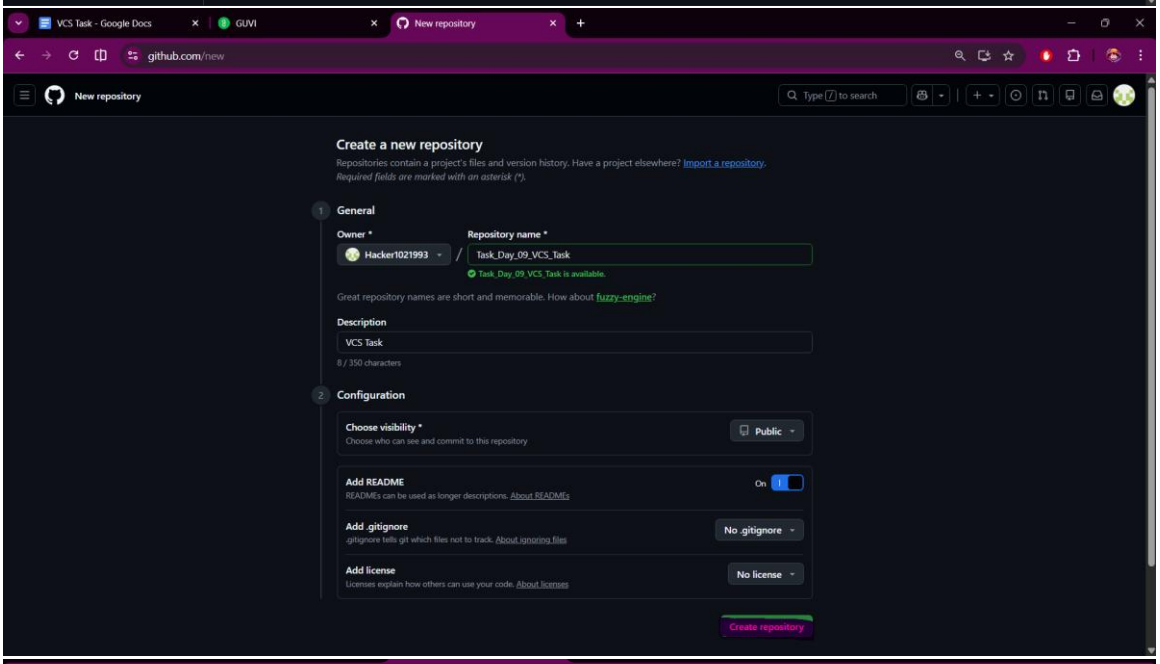
```
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ chmod +x calculator.sh
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ ./calculator.sh
Simple Calculator
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Select an operation (1-5): 1
Enter first number: 5
Enter second number: 2
Result: 7

Simple Calculator
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Select an operation (1-5): 5
Exiting...
```

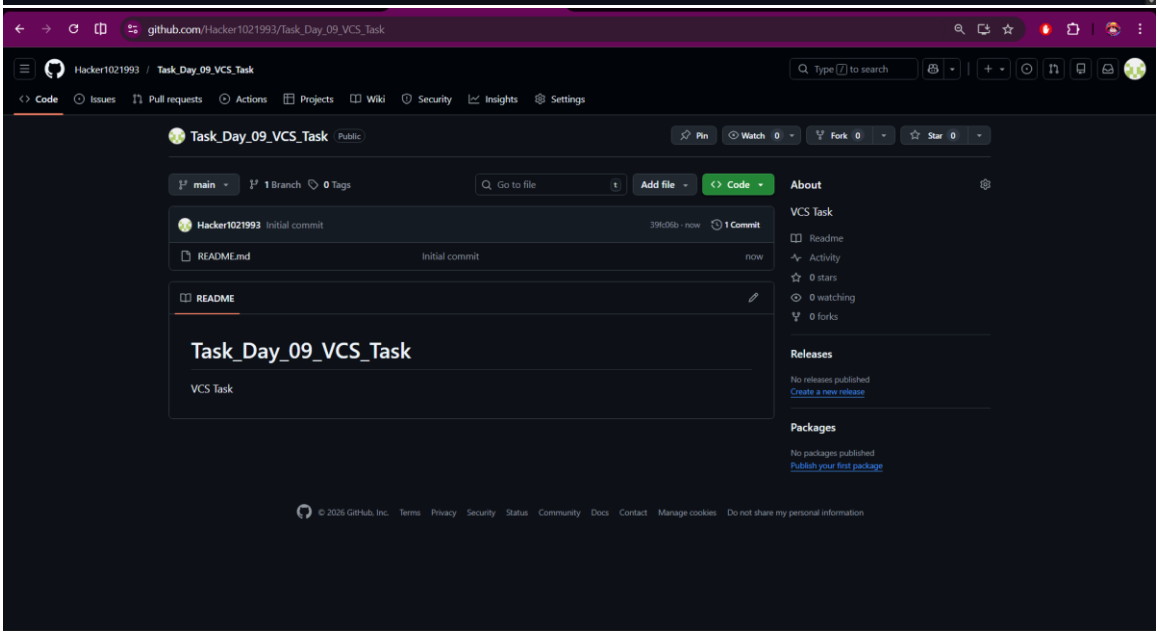
```
GHOST@GHOST: ~/Projects/I
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ ls -lah
total 16K
drwxr-xr-x  2 GHOST GHOST 4.0K Feb 20 19:52 .
drwxr-xr-x 12 GHOST GHOST 4.0K Feb 20 19:44 ..
-rwxr-xr-x  1 GHOST GHOST  836 Feb 20 19:49 calculator.sh
-rwxr-xr-x  1 GHOST GHOST  263 Feb 20 19:52 oddoreven.sh
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ ./oddoreven.sh
Enter a number: 5
5 is Odd.
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$ cat oddoreven.sh
#!/bin/bash

# Function to check if the number is odd or even
function check_odd_even {
    if (( $1 % 2 == 0 )); then
        echo "$1 is Even."
    else
        echo "$1 is Odd."
    fi
}

# Main script
read -p "Enter a number: " number
check_odd_even $number

GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_09$
```

```
GHOST@GHOST: ~/Projects/I
#!/bin/bash

# Function to display the menu
function display_menu {
    echo "Simple Calculator"
    echo "1. Addition"
    echo "2. Subtraction"
    echo "3. Multiplication"
    echo "4. Division"
    echo "5. Exit"
}

# Function to perform calculations
function calculate {
    case $1 in
        1) echo "$2 + $3" | bc ;;
        2) echo "$2 - $3" | bc ;;
        3) echo "$2 \* $3" | bc ;;
        4) echo "$2 / $3" | bc ;;
        *) echo "Invalid option" ;;
    esac
}

# Main script loop
while true; do
    display_menu
    read -p "Select an operation (1-5): " option

    if [[ "$option" -eq 5 ]]; then
        echo "Exiting..."
        break
    fi

    read -p "Enter first number: " num1
    read -p "Enter second number: " num2

    result=$(calculate "$option" "$num1" "$num2")
    echo "Result: $result"
    echo ""
done
~
~
~
"calculator.sh" 40L, 836B
```

3. **Initiating an empty repository on GitHub:**
   Creating a new repository in GitHub, as I am already maintaining a repository for Tasks, I will be using that but This is just to show how to create a new repository. [I won't be using this repo for this task]







4. **Converting the local directory into a GitHub Repository:**
   Initializing git in the local directory to convert into a GitHub Repository

```
GHOST@GHOST: ~/Projects/[

GHOST@GHOST:~/Projects/DevOps_Task$ ll
total 52
drwxr-xr-x 12 GHOST GHOST 4096 Feb 20 19:44 ./
drwxr-xr-x  7 GHOST GHOST 4096 Feb 10 11:09 ../
drwxr-xr-x  8 GHOST GHOST 4096 Feb 20 20:00 .git/
-rwxr-xr-x  1 GHOST GHOST  406 Feb 17 16:37 README.md*
drwxr-xr-x  4 GHOST GHOST 4096 Jan 29 17:17 Task_Day_01/
drwxr-xr-x  3 GHOST GHOST 4096 Jan 28 11:34 Task_Day_02/
drwxr-xr-x  3 GHOST GHOST 4096 Jan 29 16:43 Task_Day_03/
drwxr-xr-x  4 GHOST GHOST 4096 Feb  4 06:28 Task_Day_04/
drwxr-xr-x  2 GHOST GHOST 4096 Feb 10 10:42 Task_Day_05/
drwxr-xr-x  3 GHOST GHOST 4096 Feb 10 10:42 Task_Day_06/
drwxr-xr-x  4 GHOST GHOST 4096 Feb 17 16:37 Task_Day_07/
drwxr-xr-x  3 GHOST GHOST 4096 Feb 17 16:37 Task_Day_08/
drwxr-xr-x  2 GHOST GHOST 4096 Feb 20 19:52 Task_Day_09/
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git init
Reinitialized existing Git repository in /home/GHOST/Projects/DevOps_Task/.git/
GHOST@GHOST:~/Projects/DevOps_Task$
```

5. **Linking it to GitHub for pushing the code into the repository:**
   Linking the GitHub repository into the local repository using git remote origin:

```
GHOST@GHOST:~/Projects/DevOps_Task$ git remote add origin https://github.com/Hacker1021993/DevOps_Tasks.git
error: remote origin already exists.
GHOST@GHOST:~/Projects/DevOps_Task$
```

As I already have this repository added to the local folder, it's showing the error that the remote origin already exists.

Now to push the files inside the local repository to the GitHub Repository, I am going to check the status of the GitHub Repo using **git status** and I am going to stage the new or modified files to the stage area using "**git add new-filename/foldername" or "git add ."** (. Means the all the new or modified files in the local directory)

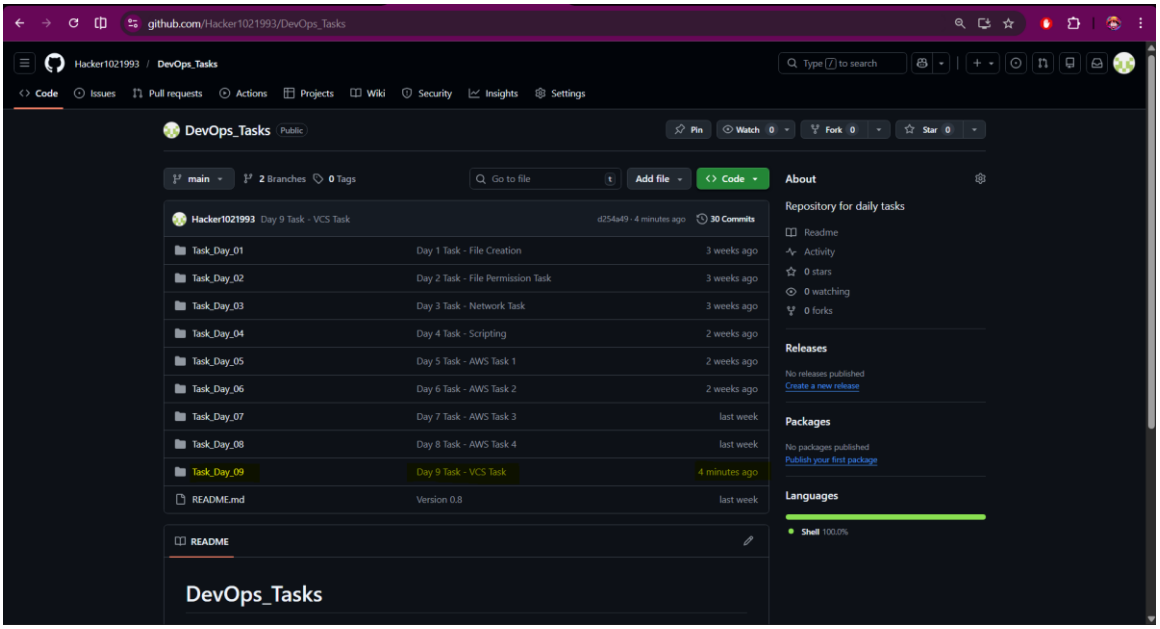
```
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Task_Day_09/

nothing added to commit but untracked files present (use "git add" to track)
GHOST@GHOST:~/Projects/DevOps_Task$ git add Task_Day_09/
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Task_Day_09/calculator.sh
        new file:   Task_Day_09/oddoreven.sh

GHOST@GHOST:~/Projects/DevOps_Task$
```

6. **Committing the change and Pushing the files to GitHub Repository:**
   Now to commit the changes made I am using **git commit -m "Commit Note"** and finally pushing the files to the GitHub Repository using **git push -u origin main**(branch name):

```
GHOST@GHOST:~/Projects/DevOps_Task$ git commit -m "Day 9 Task - VCS Task"
[main d254a49] Day 9 Task - VCS Task
 2 files changed, 55 insertions(+)
 create mode 100755 Task_Day_09/calculator.sh
 create mode 100755 Task_Day_09/oddoreven.sh
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git push -u origin main
Username for 'https://github.com': hacker1021993
Password for 'https://hacker1021993@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 939 bytes | 939.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Hacker1021993/DevOps_Tasks.git
   c2164e8..d254a49  main -> main
branch 'main' set up to track 'origin/main'.
GHOST@GHOST:~/Projects/DevOps_Task$
```

7. **Verifying the changes in the GitHub Repository:**

   As shown in the image below, the new Task_Day_09 directory has been added to the repository



8. **Performing Merge:**

Creating a separate branch to work on some tasks then merging the newly created branch with the main branch to keep the commits performed in the new branch.

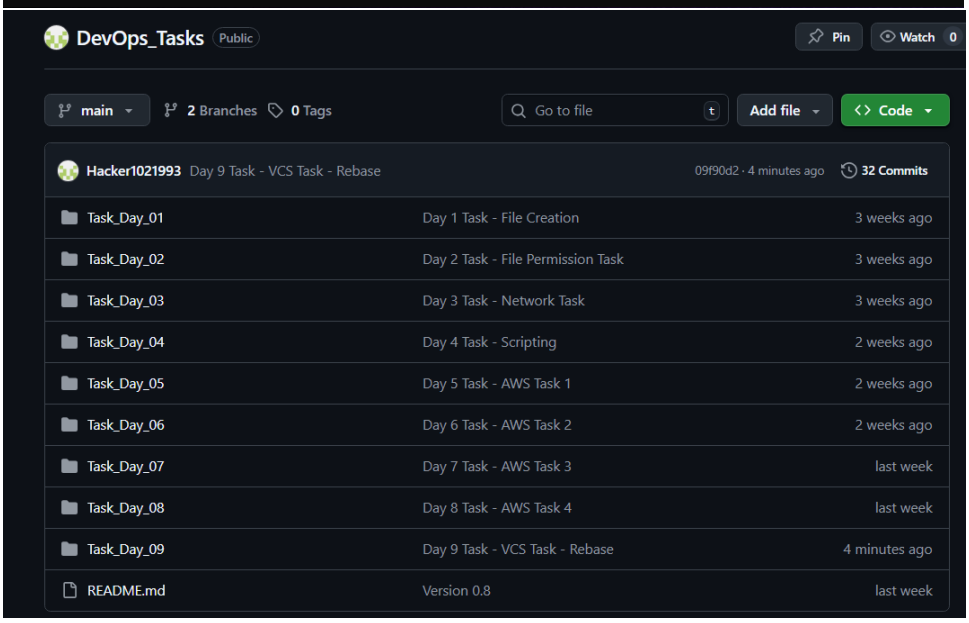
```
GHOST@GHOST: ~/Projects/[

GHOST@GHOST:~/Projects/DevOps_Task$ git branch
* main
  master
GHOST@GHOST:~/Projects/DevOps_Task$ git checkout -b vcs_task_branch
Switched to a new branch 'vcs_task_branch'
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git branch
  main
  master
* vcs_task_branch
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ touch Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ ls -l Task_Day_09
total 8
-rwxr-xr-x 1 GHOST GHOST 836 Feb 20 19:49 calculator.sh
-rwxr-xr-x 1 GHOST GHOST 263 Feb 20 19:52 oddoreven.sh
-rw-r--r-- 1 GHOST GHOST   0 Feb 20 20:38 vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch vcs_task_branch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Task_Day_09/vcstask_merge.txt

nothing added to commit but untracked files present (use "git add" to track)
GHOST@GHOST:~/Projects/DevOps_Task$ git add Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git commit -m "Day 9 Task - VCS Task - Merge"
[vcs_task_branch e3d2cd4] Day 9 Task - VCS Task - Merge
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch vcs_task_branch
nothing to commit, working tree clean
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$
```

```
GHOST@GHOST:~/Projects/DevOps_Task$ git merge vcs_task_branch
Updating d254a49..e3d2cd4
Fast-forward
 Task_Day_09/vcstask_merge.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git push -u origin main
Username for 'https://github.com': hacker1021993
Password for 'https://hacker1021993@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 411 bytes | 411.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Hacker1021993/DevOps_Tasks.git
   d254a49..e3d2cd4  main -> main
branch 'main' set up to track 'origin/main'.
GHOST@GHOST:~/Projects/DevOps_Task$
```

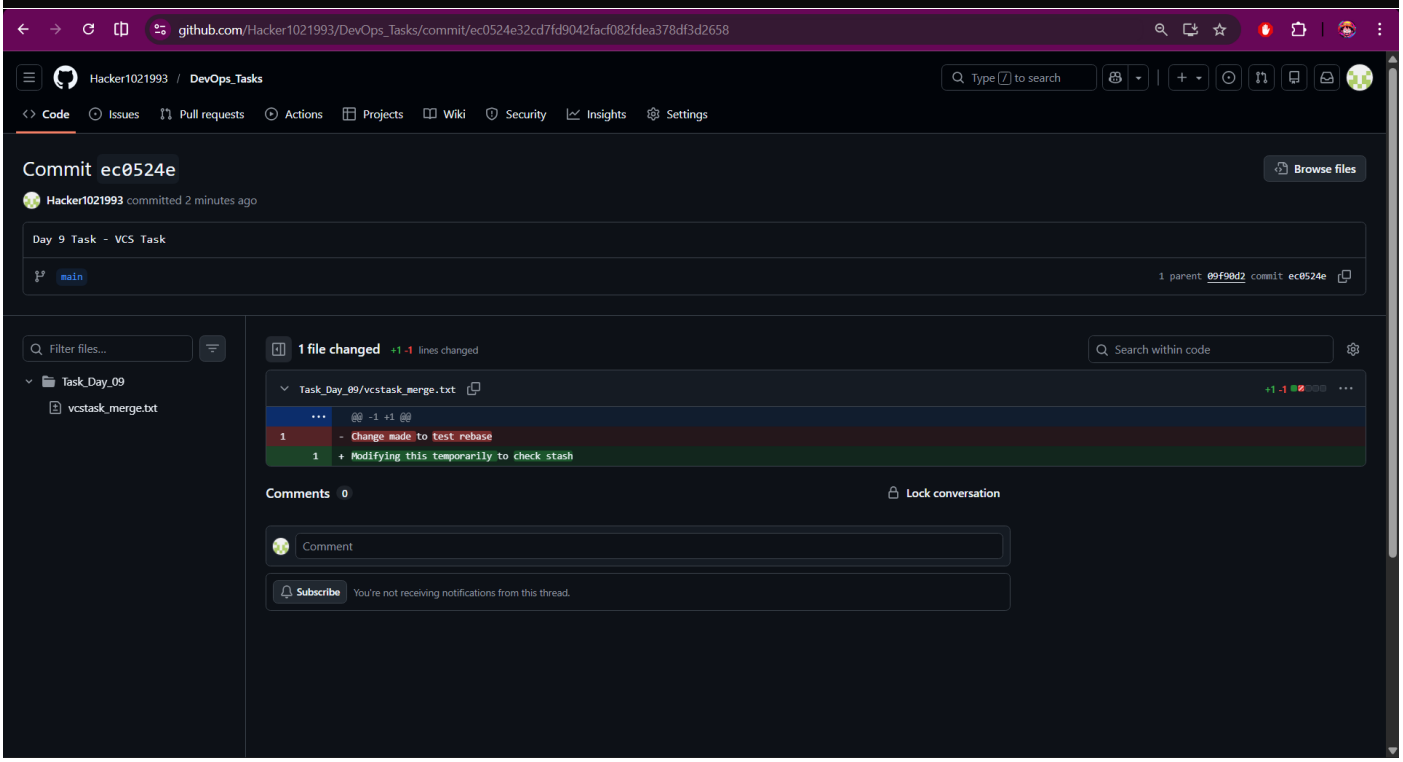
9. Rebase:

Performing Rebase command and verifying the changes in the GitHub Repository:

```
GHOST@GHOST: ~/Projects/[

GHOST@GHOST:~/Projects/DevOps_Task$ git branch
* main
  master
  vcs_task_branch
GHOST@GHOST:~/Projects/DevOps_Task$ git checkout vcs_task_branch
M       Task_Day_09/vcstask_merge.txt
Switched to branch 'vcs_task_branch'
GHOST@GHOST:~/Projects/DevOps_Task$ git branch
  main
  master
* vcs_task_branch
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ echo "Change made to test rebase" > Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$ cat Task_Day_09/vcstask_merge.txt
Change made to test rebase
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch vcs_task_branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Task_Day_09/vcstask_merge.txt

no changes added to commit (use "git add" and/or "git commit -a")
GHOST@GHOST:~/Projects/DevOps_Task$ git add Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git commit -m "Day 9 Task - VCS Task - Rebase"
[vcs_task_branch 09f90d2] Day 9 Task - VCS Task - Rebase
 1 file changed, 1 insertion(+)
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
GHOST@GHOST:~/Projects/DevOps_Task$ git branch
* main
  master
  vcs_task_branch
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git rebase vcs_task_branch
Successfully rebased and updated refs/heads/main.
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git log --online --graph
fatal: unrecognized argument: --online
GHOST@GHOST:~/Projects/DevOps_Task$ git log --oneline --graph
* 09f90d2 (HEAD -> main, vcs_task_branch) Day 9 Task - VCS Task - Rebase
* e3d2cd4 (origin/main) Day 9 Task - VCS Task - Merge
* d254a49 Day 9 Task - VCS Task
* c2164e8 Version 0.8
* 52eeec5 Day 8 Task - AWS Task 4
* 510ab51 Version 0.7
* 5ce21a1 Day 7 Task - AWS Task 3
* c18faf7 Version 0.6
* b1dd445 Day 6 Task - AWS Task 2
* c126c4c Day 6 Task - AWS Task 2
* 90052db Version 0.5
* ce433a3 Day 5 Task - AWS Task 1
* 3dc66c8 Day 4 Task - Scripting
* 9c35ccf Version 0.4
```



10. Stash:

Performing Stash to temporarily save the changes without committing:

```
GHOST@GHOST: ~/Projects/[

GHOST@GHOST:~/Projects/DevOps_Task$ ls
README.md  Task_Day_01  Task_Day_02  Task_Day_03  Task_Day_04  Task_Day_05  Task_Day_06  Task_Day_07  Task_Day_08  Task_Day_09
GHOST@GHOST:~/Projects/DevOps_Task$ echo "Modifying this temporarily to check stash" > Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$ cat Task_Day_09/vcstask_merge.txt
Modifying this temporarily to check stash
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Task_Day_09/vcstask_merge.txt

no changes added to commit (use "git add" and/or "git commit -a")
GHOST@GHOST:~/Projects/DevOps_Task$ git stash
Saved working directory and index state WIP on main: 09f90d2 Day 9 Task - VCS Task - Rebase
GHOST@GHOST:~/Projects/DevOps_Task$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
GHOST@GHOST:~/Projects/DevOps_Task$ git stash list
stash@{0}: WIP on main: 09f90d2 Day 9 Task - VCS Task - Rebase
GHOST@GHOST:~/Projects/DevOps_Task$ git stash apply
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Task_Day_09/vcstask_merge.txt

no changes added to commit (use "git add" and/or "git commit -a")
GHOST@GHOST:~/Projects/DevOps_Task$ cat Task_Day_09/vcstask_merge.txt
Modifying this temporarily to check stash
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git add Task_Day_09/vcstask_merge.txt
GHOST@GHOST:~/Projects/DevOps_Task$
GHOST@GHOST:~/Projects/DevOps_Task$ git commit -m "Day 9 Task - VCS Task"
[main ec0524e] Day 9 Task - VCS Task
 1 file changed, 1 insertion(+), 1 deletion(-)
GHOST@GHOST:~/Projects/DevOps_Task$ git push
Username for 'https://github.com': hacker1021993
Password for 'https://hacker1021993@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 373 bytes | 373.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Hacker1021993/DevOps_Tasks.git
   09f90d2..ec0524e  main -> main
GHOST@GHOST:~/Projects/DevOps_Task$
```